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Fourier Decomposition of RR Lyrae Light Curves: Practical Methods and Common Pitfalls

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Abstract. We review methods for determining periods and mean magnitudes of RR Lyrae stars from sparse, unevenly sampled photometry. We compare Lomb–Scargle, phase dispersion minimization, and multi-band periodograms, and show that intensity-averaged means from fourth-order Fourier fits are robust to within 0.01 mag for 40 or more epochs. We highlight daily aliasing as the dominant failure mode.

1. Period Search

For ground-based data with nightly cadence, aliases at $P' = P/(1 \pm P)$ are common. Inspecting the folded light curve at the top two or three peaks is strongly recommended.

2. Mean Magnitudes

Magnitude-averaged means are biased bright relative to intensity-averaged means by an amount that grows with amplitude. We recommend converting the fitted model to flux before averaging.

3. Fourier Order

Fourth-order fits capture RRab morphology adequately for typical sampling; higher orders overfit when gaps in phase coverage exceed ~0.1.

Table 1. Summary of key values.

Method	Alias rate	Mean-mag bias
Lomb–Scargle	4%	<0.01
PDM	6%	<0.01
Multi-band LS	2%	<0.01

References

Lomb, N. R. 1976, Ap&SS, 39, 447
Scargle, J. D. 1982, ApJ, 263, 835